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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 6	
TITLE: Inner and Anonymous Classes	

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

1.Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

Programme

Assignment6.java

src > Assignment6.java > Vehicle

```
1   class Vehicle {
2       String brand = "Toyota";
3       int speed = 120;
4
5       // Inner class
6       class VehicleDetails {
7           void displayDetails() {
8               System.out.println("Brand: " + brand);
9               System.out.println("Speed: " + speed + " km/h");
10          }
11      }
12
        Run | Debug
13      public static void main(String[] args) {
14
15          Vehicle v = new Vehicle();
16
17          VehicleDetails details = v.new VehicleDetails();
18          details.displayDetails();
19
20          // Anonymous class
21          Runnable action = new Runnable() {
22              public void run() {
23                  System.out.println(x: "Vehicle is moving.");
24              }
25          };
26
27          action.run();
28      }
29  }
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac Assignment6.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java Vehicle
Brand: Toyota
Speed: 120 km/h
Vehicle is moving.
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> S
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

Programme

Assignment6.java

src > Assignment6.java > FoodDelivery

```java
class FoodDelivery {

    String customerName = "Riya";
    String foodItem = "Pizza";

    // Inner class
    class OrderDetails {
        void displayOrder() {
            System.out.println("Customer: " + customerName);
            System.out.println("Food Item: " + foodItem);
        }
    }

    Run | Debug
    public static void main(String[] args) {

        FoodDelivery order = new FoodDelivery();

        OrderDetails details = order.new OrderDetails();
        details.displayOrder();

        // Anonymous class
        Runnable deliveryStatus = new Runnable() {
            public void run() {
                System.out.println(x: "Delivery Status: Order is out for delivery.");
            }
        };

        deliveryStatus.run();
    }
}
```

Output

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Debug: FoodDelivery

```
PS C:\Users\RIKISHA\PROGRAMMING\Java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt
_socket,server=n,suspend=y,address=localhost:58763' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp
' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'FoodDelivery'
Customer: Riya
Food Item: Pizza
Delivery Status: Order is out for delivery.
PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

This assignment helped us understand the concept of Inner Classes and Anonymous Classes in Java. We learned how to create and use both types of classes and

understand their practical applications. These classes help improve code organization, encapsulation, flexibility, and readability in Java programs.

GitHub Screenshot